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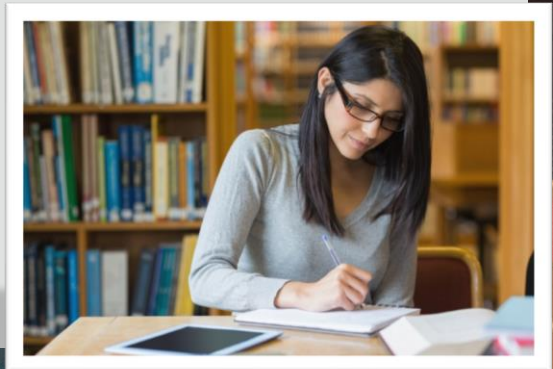
Academy

Database Foundations

6-1

Introduction to Oracle Application Express

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Objectives

- This lesson covers the following objectives:
 - Distinguish between application software and system software and give an example of each
 - Log-in to the Oracle Application Express practice environment
 - Execute a simple query to retrieve information from the Database
 - Apply the rules of SQL to display all columns and a subset of columns specified by criteria



Purpose

- Every day, in one way or another, we come in contact with computer applications
 - If you checked your email today, it was probably done using an application
 - If you bought an item at a grocery store, the clerk scanned the item using an application that calculated your bill and updated the store inventory
- In this course, you will learn the syntax of SQL using the application called Oracle Application Express

Application Programs

- Although computers have been around for a very long time (possibly before you were born), their use for business and personal computing didn't take place until application software programs were developed
- Application programs allowed the end user—people like you and me—to be able to buy fully developed, ready-to-use programs. It was no longer necessary to know how the program worked, just that it did work and accomplished what we wanted it to do

Application program software is different from system software. System software consists of low-level programs designed to interact with the computer hardware. Operating systems, compilers, and system utilities are examples of system software. In contrast, application software includes programs for word processing, databases, gaming, email, and graphics.

Application Programs

- Yahoo.com uses the Oracle database to store data
- Rather than having everyone who wants to search the database or retrieve email learn SQL, an application has all of the SQL (and other coding languages) pre-programmed into it
- With a few mouse clicks, users have access to all of the information they need

Using Applications

- An application is like a car
 - To drive a car, you need to know enough to make it work.
 - It has a friendly “shell” to hide all the things that you don’t need to know, such as how the transmission works or how fuel like petrol or diesel is used to power the engine
- Could you ever get your driver's license if you had to demonstrate an understanding of every system—electrical, powertrain, hydraulic, fuel, etc.—used to make the car run?

Oracle Application Express

- Oracle Application Express is a web application development, deployment, and maintenance tool



Oracle APEX

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DFo 6-1
Introduction to Oracle Application Express

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- Oracle Application Express is a web-based development and deployment tool that is available with the Oracle database. It enables you to create database-centric web applications that are reliable, scalable, and secure. It has several built-in features and wizards that quicken your development process. Some of the key features are listed in the slide.
- The tool has a user-friendly graphical interface. Using Oracle Application Express requires minimal programming knowledge.

Note: In this course, you will use Oracle Application Express.

Oracle Application Express Components

- Oracle Application Express has these components:
 - SQL Workshop
 - Application Builder
 - Object Browser
- To learn SQL, you will use the SQL Workshop component
- To design an application, you use Application Builder
- For instruction on using APEX please refer to the APEX iAcademy Learner Guide in the Member Hub

Project Exercise

- DFo_6_1_Project
 - Oracle Baseball League Store Database
 - Upload and run a script in APEX



Basic SELECT Statement

- The SELECT * command returns all the rows in a table
 - The syntax is:

```
SELECT *  
FROM <table name>;
```

- For example:

```
SELECT *  
FROM employees;
```

Syntax: The rules governing the formation of statements in a programming language.

Students should enter the above example into APEX to see the results output.

SELECT Statement with a Condition

- To return a subset of the data, modify the SELECT statement.
 - The syntax is:

```
SELECT <column name 1, column name 2, etc.>  
FROM <table name>  
WHERE <condition>;
```

- For example:

```
SELECT first_name, last_name, job_id  
FROM employees  
WHERE job_id = 'SA_REP';
```

The <condition> starts with the word WHERE followed by a <column name> and a Comparison Operator (=, >, <, etc.) followed by a value or IS NULL, IS NOT NULL.

Correcting errors

- When entering SQL commands, it is important to use the correct spelling, otherwise you will get an error message
- For example (SELECT: spelling incorrect):
- Would result in the error message:

```
SEECT *  
FROM employees;
```

- To rectify, simply correct the spelling and run again

```
ORA-00900: invalid SQL statement
```

Terminology

- Key terms used in this lesson included:
 - Application software
 - System software
 - Oracle Application Express
 - Syntax
 - Subset
 - Comparison Operator

Summary

- In this lesson, you should have learned how to:
 - Distinguish between application software and system software and give an example of each
 - Log-in to the Oracle Application Express practice environment
 - Execute a simple query to retrieve information from the Database
 - Apply the rules of SQL to display all columns and a subset of columns specified by criteria



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